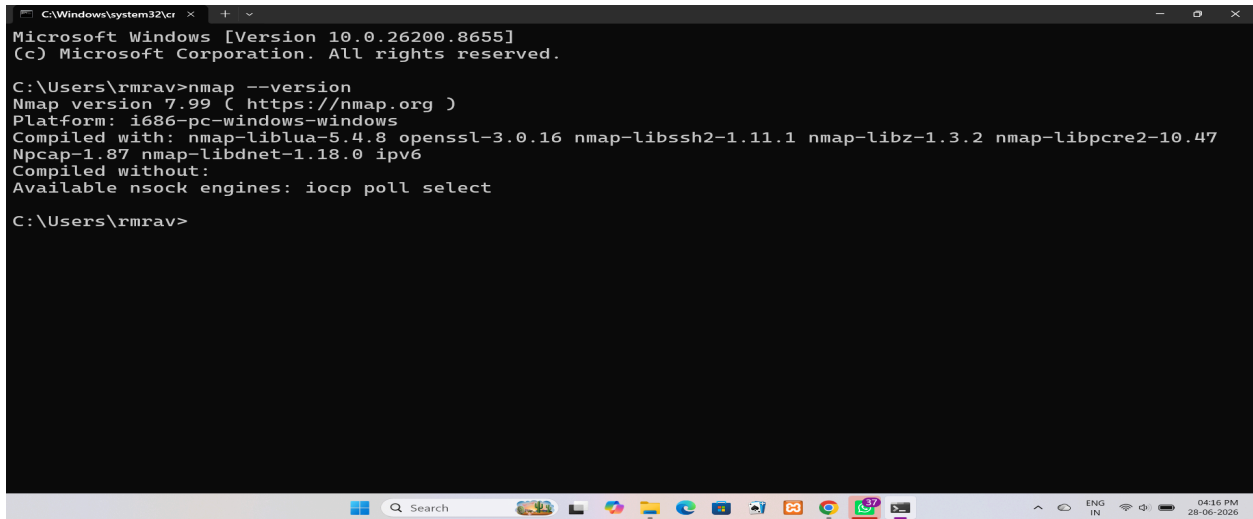


PART A:-

Nmap was successfully installed on the system. The installation was verified by executing the following command:

```
nmap --version
```



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rmrav>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47
Npcap-1.87 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\rmrav>
```

Part B: Scan Your Local Machine

Command Used

```
nmap localhost
```

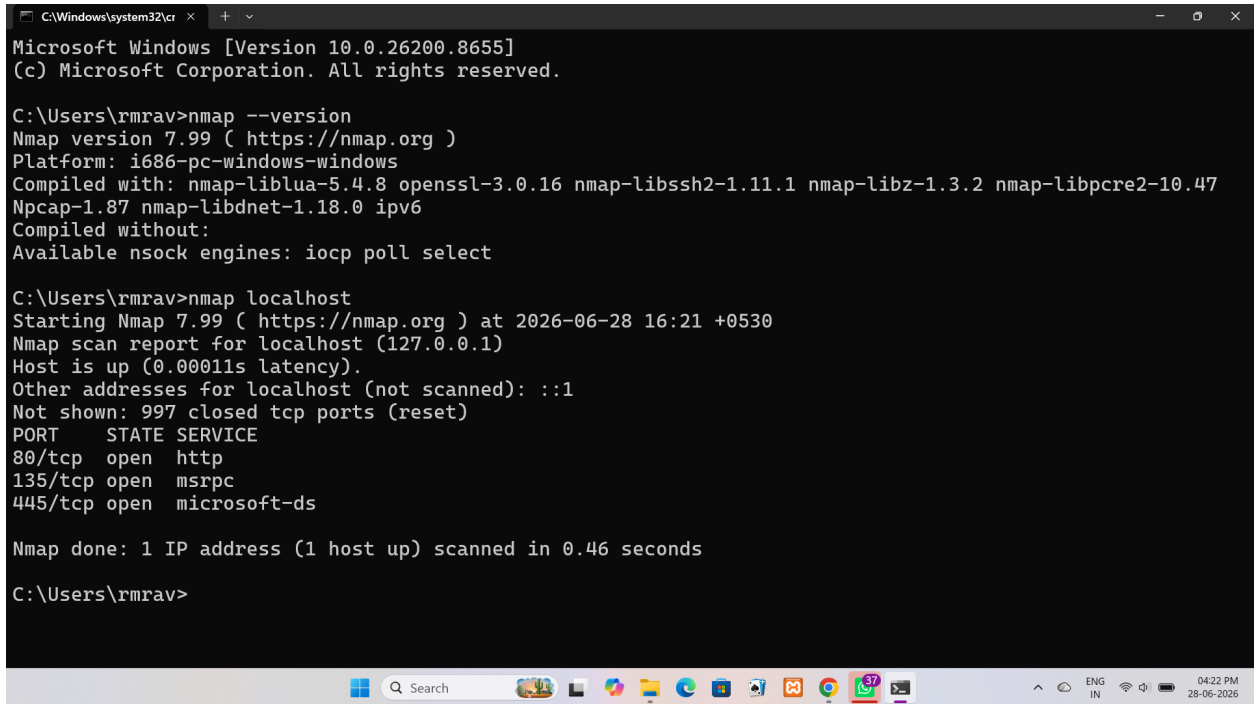
Scan Results

Total Open Ports: 3

Port Number	Service Running
80/tcp	http
135/tcp	msrpc
445/tcp	microsoft-ds

Result

The Nmap scan was successfully performed on the local machine (**localhost**). The scan detected **3 open TCP ports: 80 (HTTP), 135 (MSRPC), and 445 (Microsoft-DS)**, indicating that these services were active on the local system.



```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rmrav>nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.11.1 nmap-libz-1.3.2 nmap-libpcap-1.0.4
Npcap-1.87 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\rmrav>nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 16:21 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.46 seconds

C:\Users\rmrav>
```

Part C: Service Version Detection

Command Used

```
nmap -sV localhost
```

Service Version Detection

Port Number	Service Name	Version Detected
80/tcp	HTTP	Microsoft IIS httpd 10.0
135/tcp	MSRPC	Microsoft Windows RPC
445/tcp	Microsoft-DS	Version not detected

Result

The `nmap -sV localhost` command was successfully executed on the local machine. Nmap detected the service versions for the HTTP and MSRPC services. The version information for the Microsoft-DS service could not be determined.

```

C:\Windows\system32\cmd.exe
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
445/tcp   open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 0.46 seconds

C:\Users\rmrav>nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 16:24 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00011s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Microsoft IIS httpd 10.0
135/tcp   open  msrpc       Microsoft Windows RPC
445/tcp   open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.31 seconds

C:\Users\rmrav>

```

Part D: Operating System Detection

Command Used

`nmap -O localhost`

Was the operating system detected?

Answer: Yes

Which OS was identified?

Answer: Microsoft Windows 11 (24H2 – 25H2)

Why is OS detection useful during penetration testing?

Operating system detection helps penetration testers identify the target operating system. This information is useful for selecting appropriate vulnerability scans, identifying potential security weaknesses, choosing compatible exploits, and planning effective security assessments.

```
C:\Windows\system32\cmd.exe
135/tcp open  msrpc      Microsoft Windows RPC
445/tcp open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.31 seconds

C:\Users\rmrav>nmap -O localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-28 16:26 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0074s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 - 25H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.59 seconds

C:\Users\rmrav>
```

Part E: Common Port Research

Port	Protocol	Purpose	Common Security Risks
------	----------	---------	-----------------------

20/21	FTP (File Transfer Protocol)	Transfers files between a client and a server. Port 20 is used for data transfer, while Port 21 is used for control commands.	Data and passwords are transmitted in plain text, making them vulnerable to interception and brute-force attacks.
22	SSH (Secure Shell)	Provides secure remote login and encrypted communication between systems.	Weak passwords, brute-force attacks, and outdated SSH software can lead to unauthorized access.
23	Telnet	Allows remote login to a computer or network device.	Sends all data, including usernames and passwords, in plain text, making it highly insecure.
25	SMTP (Simple Mail Transfer Protocol)	Sends email between mail servers and from clients to mail servers.	Email spoofing, spam relay, and phishing attacks if the server is not properly configured.
53	DNS (Domain Name System)	Translates domain names into IP addresses.	DNS spoofing, cache poisoning, and DNS amplification attacks.
80	HTTP (Hypertext Transfer Protocol)	Delivers web pages over the internet without encryption.	Data interception, man-in-the-middle attacks, and web application vulnerabilities.
110	POP3 (Post Office Protocol v3)	Retrieves email from a mail server to a local device.	Credentials may be exposed if encryption is not used; susceptible to eavesdropping.

143	IMAP (Internet Message Access Protocol)	Allows users to access and manage email stored on a mail server.	Unencrypted connections can expose email data and login credentials.
443	HTTPS (Hypertext Transfer Protocol Secure)	Provides encrypted communication for secure web browsing.	Weak SSL/TLS configurations, expired certificates, and protocol vulnerabilities.
445	SMB (Server Message Block)	Enables file and printer sharing on Windows networks.	Vulnerable to malware, ransomware (such as WannaCry), and unauthorized file access if not secured.
3389	RDP (Remote Desktop Protocol)	Allows remote access and control of Windows computers.	Brute-force attacks, credential theft, and unauthorized remote access if exposed to the internet.

Part F: Scan Analysis

1. Which services are currently running?

The Nmap scan detected the following services running on the local machine:

- Port 80: HTTP (Microsoft IIS Web Server)
- Port 135: Microsoft RPC (Remote Procedure Call)
- Port 445: Microsoft-DS (SMB – Server Message Block)

2. Are all open ports necessary?

Not necessarily. Some ports are required for the operating system or installed applications to function correctly. However, services that are not being used should be disabled to reduce the system's attack surface.

3. Which services could become security risks if misconfigured?

The HTTP (Port 80) service could expose web applications to attacks if it is not properly configured. Microsoft RPC (Port 135) and SMB (Port 445) can also become security risks if they are exposed to untrusted networks or left unpatched, as they have been targeted by malware and unauthorized access attempts.

4. Which port would you disable if it wasn't required?

If file and printer sharing is not needed, I would disable Port 445 (SMB) because it has been associated with several security vulnerabilities and malware attacks. Disabling unused services helps improve system security.

Part G: Nmap Scan Report

Scan Report

Scan Date: 28 June 2026

Target System: localhost (127.0.0.1)

Commands Used

`nmap --version`

`nmap localhost`

`nmap -sV localhost`

`nmap -O localhost`

Open Ports

Port	State
------	-------

80/tcp	Open
--------	------

135/tcp	Open
---------	------

445/tcp	Open
---------	------

Running Services

Port	Service	Version
------	---------	---------

80/tcp	HTTP	Microsoft IIS httpd 10.0
--------	------	--------------------------

135/tcp	MSRPC	Microsoft Windows RPC
---------	-------	-----------------------

445/tcp	Microsoft-DS (SMB)	Version Not Detected
---------	--------------------	----------------------

Operating System

- **Operating System:** Microsoft Windows 11 (24H2 – 25H2)

Observations

- The target system was reachable and responded successfully to all Nmap scans.
- Three TCP ports (80, 135, and 445) were found to be open.
- The HTTP service was running on Microsoft IIS 10.0.
- Microsoft Windows RPC and SMB services were also active.
- Nmap successfully detected the operating system as Microsoft Windows 11.

Recommendations

1. Disable unnecessary services and close unused ports.
2. Keep Windows and all services updated with the latest security patches.
3. Restrict access to SMB (Port 445) and RPC (Port 135) using a firewall.
4. Regularly monitor open ports and network services.
5. Use strong authentication and security configurations for all running services.

Conclusion

During this practical, I learned how to use Nmap to perform basic network reconnaissance and analyze a local system. I verified the installation of Nmap, scanned my computer to identify open ports, detected running services, and performed operating system detection. The scan results showed that ports **80 (HTTP)**, **135 (MSRPC)**, and **445 (SMB)** were open. Nmap also identified the operating system as **Microsoft Windows 11 (24H2–25H2)**. By using the **-sV** option, I was able to detect service versions, including Microsoft IIS 10.0 running on the HTTP service. This activity demonstrated how network scanning helps identify active services and potential attack surfaces on a system. I also learned that unnecessary open ports and improperly configured services can increase security risks. Regular scanning, timely software updates, proper firewall configuration, and disabling unused services are important practices for improving system security. Overall, this exercise enhanced my understanding of reconnaissance techniques, port scanning, service identification, and operating system detection, which are essential skills in cybersecurity and penetration testing.